Fiscal Year:	FY 2022	**Task Last Updated:**	FY 03/07/2022
PI Name:	Matson, Douglas Ph.D.		
Project Title:	Round Robin - Thermophysical Property Measurement		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Medford	**State:**	MA
Zip Code:	02155	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	04/01/2017	**End Date:**	02/29/2024
No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:	6	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Sansoucie, Michael	**Contact Phone:**	256.544.5269
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 02/29/2024 per PI (Ed., 4/4/22).		
Key Personnel Changes/Previous PI:	CoInvestigator Mikhail Krivilev from Udmurt State University has been dropped from the program due to inactivity on the grant. He was unpaid by NASA since he is supported by ROSCOSMO so no fiscal impact.		
COI Name (Institution):			
Grant/Contract No.:	NNX17AH41G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Researchers go to space to conduct material property measurements that are required to model terrestrial manufacturing processes to provide higher performance and discover new materials to make people's lives better around the world. We justify the expense of doing tests in microgravity by concentrating on those materials which are impossible to evaluate on Earth or have properties that can only be measured with higher precision when gravitational accelerations are minimized. We know we get better results but we do not yet know how much better. The JAXA (Japanese Aerospace Exploration Agency) Electrostatic Levitation Furnace (ELF) was designed to process high temperature oxides but it can be readily adapted to allow testing of industrially important metals and alloys as well. By selecting materials that are already being processed using another technique, the European Space Agency (ESA) Electromagnetic Levitation facility on the International Space Station (ISS), these two facilities can be compared so that researchers know their relative strengths such that they may complement each other to maximize science output from space tests. Our objectives are to conduct space experiments to compare to ground results on a baseline material set and to model the influence of gravity on measurement techniques by various research groups across the world. Our approach is to employ Round Robin testing by multiple researchers on the same material using different techniques, on different facilities, with different analysis protocols to compare results. The significance of this work is that we will then know the relative increase in measurement fidelity and which material works best in which facility to efficiently use the limited available space sciences resources such as launch cost, crew time, and telemetry bandwidth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This program leverages the existing ESA International Topical Team organization to develop a strategy to compare the performance of current flight and ground facilities investigating thermophysical property measurement using levitation techniques. New experiments are proposed for the JAXA-ELF facility on the ISS to complement other ongoing investigations that use both the ISS-based Materials Science Laboratory-Electromagnetic Levitator (MSL-EML) and ground-based facilities. We will efficiently use available resources, both fiscal and crew time, by selecting materials compatible with those previously prioritized by other existing research groups. In doing so, we will expand the capabilities of space electrostatic levitation processing to allow for investigation of additional thermophysical material properties and manifesting new material classes across microgravity platforms. The key scientific questions we address are: (1) How do we control and minimize gravity-induced systematic error in thermophysical property measurement and does theory explain the observed deviations in results? (2) For which material classes are space-based platforms delivering thermophysical property measurement results with higher accuracy and/or precision than ground-based facilities? Thermophysical property testing in microgravity provides enhanced fidelity of results purportedly due to three main effects. First, without strong gravitational accelerations and with reduced levitation forces a more spherical sample allows for better analysis of experimental behavior. Second, better control of convection in space results in higher measurement precision. Third, sedimentation and buoyancy induced segregation are eliminated in reduced gravity. But how much better do we do in space as compared to ground? The obvious solution is to quantify variability across platforms – both in space and on ground.
Task Progress:	• Graduated 3 PhD and 1 MS students. • Published 2 peer-reviewed articles & 8 presentations (see list at end of document). • Selected as Chair of the International Space Station National Laboratory (ISS-NL) User Advisory Committee supporting Center for the Advancement of Science in Space (CASIS) Ops. • Selected as Co-chair of the National Academies Committee on Biological and Physical Sciences in Space to oversee the 2022 NASA Decadal Study. • Conducted flight testing during International Space Station (ISS) Expedition 64 using the Japan Aerospace Exploration Agency (JAXA)electrostatic levitation furnace (ELF) facility. Successfully finished Sample holder S/N 008, preparing Expedition 68 samples. • Attended the 37th Annual American Society for Gravitational and Space Research (ASGSR-37) 2021 Annual Conference, November 3-6 as organizer, and recipient of Fellow Award. • Featured speaking engagements at: International Space Station Research and Development (ISSR&D) Plenary Panel, NASA Headquarters (HQ) Kibo Utilization Symposium, Japan Microgravity Application Society Conference (JASMAC-33), and the United Nations Office for Outer Space Affairs Webinar #5 - Physical Sciences: Materials Science.
Bibliography Type:	Description: (Last Updated: 05/05/2026)
Articles in Peer-reviewed Journals	Jeon S, Liu X, Azersky C, Ren J, Zhang S, Chen W, Hyers RW, Costa K, Kolbe M, Matson DM. "Particle size effects on dislocation density, microstructure, and phase transformation for high-entropy alloy powders." Materialia. 2021 Jul 13;18:101161. https://doi.org/10.1016/j.mtla.2021.101161 , Jul-2021
Articles in Peer-reviewed Journals	Matson DM, Liu X, Rodriguez JE, Jeon S, Shuleshova O. "Retained free energy with enhanced nucleation during electrostatic levitation of undercooled Fe-Co alloys." Crystals. 2021 Jun 24;11(7):730. https://doi.org/10.3390/cryst11070730 , Jun-2021
Awards	Matson DM. "American Society for Gravitational and Space Research (ASGSR) Fellow, November 2021." Nov-2021
Dissertations and Theses	Nawer J. "Thermophysical property measurement using containerless techniques in microgravity and terrestrial environment." Dissertation, Tufts University, January 2022. , Jan-2022
Dissertations and Theses	Bracker G. Coadvised with R.W. Hyers. "Solidification experiments and magnetohydrodynamic models in electromagnetic levitation." Dissertation, University of Massachusetts-Amherst, November 2021. , Nov-2021
Dissertations and Theses	Leitner T. Coadvised with G. Pottlacher. "Thermophysical property measurement of industrial metals and alloys using electromagnetic levitation." Dissertation, Graz University of Technology, May 2021. , May-2021